Development of a Hydrophobic Vapor-Deposition Coating Process for Test Strips

1. Scope of Application

This hydrophobic coating method is intended only for the surface and bulk hydrophobization of fiber-based substrates with compatible physicochemical properties, including standard qualitative filter paper, thin- and thick-grade glass-fiber test strips, cellulose acetate test strips, and related materials. Performance under extreme conditions, including high temperature, radiation, and exposure to organic solvents, has not been evaluated. The limitations of applicability should therefore be noted.

2. Rationale and Objectives for Hydrophobizing the Test Strip

(1) The target product is a freeze-dried test strip carrying viable yeast. Because the design is intended to support both qualitative colorimetric readout and quantitative analysis, the viable-yeast growth zones should form regular geometric patterns with sharp, well-defined boundaries to facilitate visual inspection and subsequent image acquisition, recognition, and analysis.

(2) The vacuum-assisted spotting method developed in parallel does not adequately resolve the problems of inconsistent spot size and poorly defined edges caused by liquid spreading and penetration during test-strip preparation and yeast reactivation and growth testing.

(3) During the construction of other experimental setups, the volatile adhesive ergo.5300 was inadvertently found to produce a hydrophobic coating effect on surrounding materials. The team observed this phenomenon and subsequently developed the present method based on this observation.

3. Reagents

(1) Copper Sulfate Reagent (100 mL)

Component	Amount
$CuSO_4 \cdot 5H_2O$	4.99 g
Ultrapure water	Up to 100 ml

Table 1. Copper Sulfate Reagent

Copper sulfate is used to impart color to the water for evaluation of the hydrophobicity of the test strips. At this concentration, the color is sufficiently saturated and provides the best visual appearance in photographs.

(2) ergo.® 5300 Instant Adhesive

This is a single-component, room-temperature, fast-curing α-cyanoacrylate-based adhesive that can bond rubber, plastics, metals, and other materials. It cures rapidly in the presence of moisture on material surfaces and has a certain degree of volatility.

4. Materials

(1) Qualitative Filter Paper

Composed of cotton fibers, with good overall physical properties, relatively high liquid-holding capacity, and good toughness; it is easy to cut into and divide into various shapes.

(2) CA Cellulose Acetate Filter Membrane (Aqueous-System Type)

Pore size: 0.45 μm.

It has good uniformity but somewhat poorer mechanical properties. The material readily develops electrostatic charge and may undergo physical adhesion with similar filter paper and with container walls; it is also relatively fragile.

(3) Whatman™ GF/B Glass Fiber Filter Paper

Pore size: 1.0 μm.

Thickness: 675 μm.

It has good mechanical properties and high liquid-holding capacity, but relatively low toughness. Because it is composed of glass fibers, basic protective measures should be taken during cutting to minimize the release of airborne particulate matter.

(4) Whatman™ GF/A Glass Fiber Filter Paper

Pore size: 1.6 μm.

Thickness: 260 μm.

It has good mechanical properties and relatively high liquid-holding capacity, but relatively low toughness. Because it is composed of glass fibers, basic protective measures should be taken during cutting to minimize the release of airborne particulate matter.

5 Protocol

(1) Assembly of the Vapor-Deposition Setup

- Use a plastic Petri dish as the base. Cut the caps from 8-strip PCR tubes as required, dividing them into individual cap units or continuous sections of different lengths.
- According to the dimensions of the test strips to be treated and the expected number of strips to be treated simultaneously in a single dish, affix the PCR tube caps to the base using fast-setting adhesive. Position the test strips securely on the caps, ensuring that they remain stable while exposing as much of their surface area as possible to the vapor-phase coating agent.
- Place mock test strips on the support and gently shake the setup to assess placement stability, then adjust the spacing of the caps as needed. Close the Petri dish lid to verify that the lid does not contact the filter paper, and make further fine adjustments if necessary.

(2) Optimization of Process Conditions

- Design different experimental groups using a controlled-variable approach, varying the test-strip material, test-strip dimensions, vapor-deposition time, adhesive amount, and number of strips treated per group. The test-strip materials include qualitative filter paper, CA cellulose acetate filter membrane, Whatman GF/B glass fiber filter paper, and Whatman GF/A glass fiber filter paper. Test-strip dimensions range from 5 mm×5 mm to 10 mm×10 mm. Vapor-deposition time ranges from 5 min to 60 min. The adhesive amount ranges from 1 to 3 drops.
- Add ergo.5300 adhesive dropwise to the adhesive application area, then seal the setup for the planned vapor-deposition period.

(3) Exploration of Hydrophobic Pattern Formation

- Design hydrophilic and hydrophobic regions on the test strip according to the intended pattern.
- Method 1 (adhesive-mask method): Using laboratory adhesive label stickers, cut them into the designed shapes of the hydrophilic regions on the test strips. Prepare two identical pieces for

each sample, remove the backing paper, and apply them to the upper and lower surfaces of the test strip, respectively, with the two pieces aligned with each other.

- Method 2 (pressure-mask method): Fabricate a plastic mold matching the shape of the hydrophilic region. The mold should have a smooth and flat surface. Likewise, prepare two identical pieces for each sample, place them against the upper and lower surfaces of the test strip, respectively, and clamp them tightly using a woodworking clamp.

*Note: Samples prepared according to Section **5.(3)** must still undergo the treatment described in Section **5.(2)** before proceeding to the next step.*

(4) Evaluation of Hydrophobic Performance

- Take a clean plastic Petri dish and add an appropriate amount of copper sulfate solution. Immerse the test strips from each experimental group in the solution and observe the resulting behavior.
- Observe whether the test strips float or sink. Allow them to stand for a period of time and then observe them again.
- Observe how the test strips are wetted by the solution. Allow them to stand for a period of time and observe them again, paying particular attention to whether the wetted pattern matches the mold geometry and whether the boundaries remain clearly defined.